

Exploring Aligned and Disaligned Multimodal Features in Recommender Systems

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Outline

- 1 Introduction
- 2 Aligned Models
- 3 Experimental Settings
- 4 Experiments
- 5 Discussion & Conclusion

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Recommender Systems

- Guide users to relevant items in a large space [5]
- Approaches: Content-Based, Collaborative, Hybrid, Knowledge-Based, Context-Aware

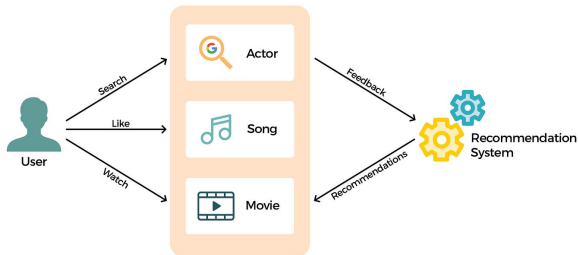


Figure: Example of a Recommender System

Multimodal Recommender Systems

- Use multiple modalities (text, image, audio) to enrich item representations [7]
- Encoding options: concat, fusion/attention, GNNs [1, 4]



Figure: Example of multimodal data for a movie (Star Wars)

Motivation & Problem Statement

Traditionally, MMRec systems extract features using independent encoders (e.g., BERT for text, ResNet for images). In other domains, like Few-Shot Learning for Image Classification, aligned features (CLIP) have shown superior performance with respect to disaligned ones.

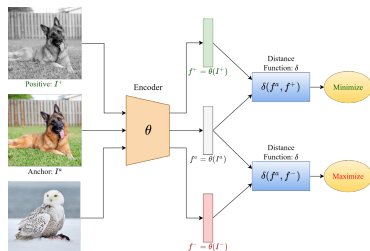
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Contrastive Learning

- Self-supervised method: learns representations that bring similar samples closer [3]
- Goal:** Minimize distance between positive pairs, maximize distance between negative pairs.
- Contrastive loss:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} [k \neq i] \exp(\text{sim}(z_i, z_k)/\tau)}$$



CLIP (Contrastive Language-Image Pre-training)

- Aligns image and text embeddings in a shared space (OpenAI) [6]
- Relevance:** Large multimodal encoders like CLIP can improve performance when fine-tuned end-to-end [8]

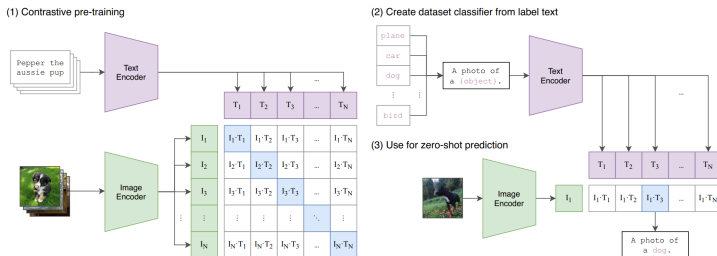


Figure: CLIP aligns visual and textual concepts

AudioCLIP

- Extends CLIP by adding an **audio encoder** (ESResNeXt) [2]
- **Training process:**
 - 1 Audio-head pre-training
 - 2 Cooperative distillation (aligning audio to frozen CLIP space)
 - 3 Full tri-modal fine-tuning on AudioSet

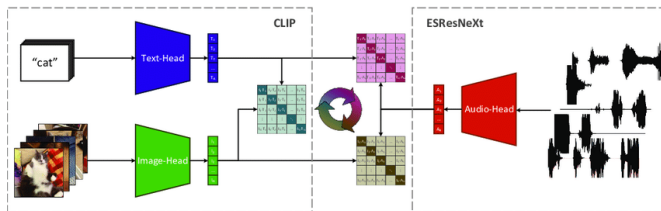
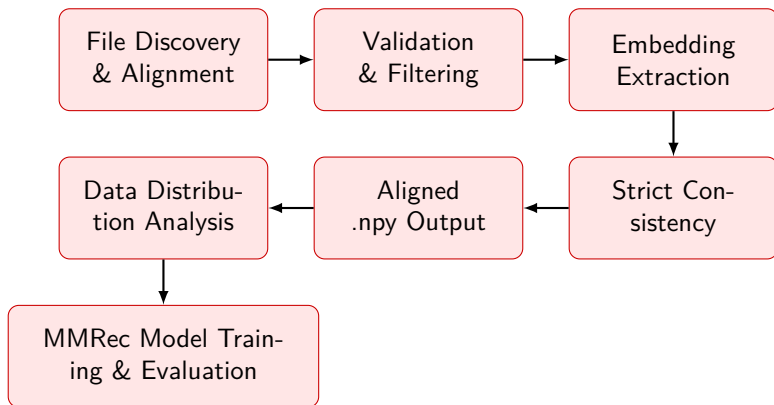


Figure: Tri-modal alignment in AudioCLIP

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Pipeline



Research Questions

RQ1: Qualitative Analysis (Geometry)

How does the alignment process (e.g., CLIP, AudioCLIP) affect the geometric distribution of the feature space?

RQ2: Quantitative Analysis (Performance)

Can the integration of *aligned* multimodal embeddings (text, image, audio) enhance recommendation accuracy compared to unimodal or non-aligned approaches?

Datasets Overview

Challenge: Dealing with different structural levels of multimodal data.

Feature	MovieLens 1M	LastFM
Domain	Movies	Music
Interactions	User-Movie (Explicit Rating)	User-Artist (Implicit Listen)
MM Level	Item Level (Movie)	Track Level (Sub-Item)
Modalities	Posters, Trailers, Plots	Audio, Covers, Metadata
Pipeline	Direct Mapping	Artist-to-Track Expansion

Table: Comparison of datasets and processing strategies

Models Evaluated

We compared performance across simpler projection models and complex Graph Neural Networks:

- **BPR (Baseline):** Bayesian Personalized Ranking (ID-only).
- **VBPR:** Visual BPR extended to support multimodal features (projection-based).
- **LATTICE:** GNN that mines latent structures between items.
- **FREEDOM:** GNN with modality-specific encoders and attention fusion.

Metrics Evaluated:

- Recall@k
- NDCG@k
- Precision@k
- MAP@k

Experimental Settings:

- **Cut-offs (k):**
 $k \in \{5, 10, 20, 50\}$
- **Filtering:** 5-core filtering applied to ensure interaction density.

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Embedding Distribution Analysis (UMAP + KDE)

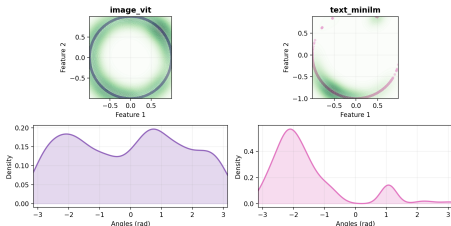
To understand *why* models perform differently, we analyzed the geometric structure of the embeddings.

- **Technique:** UMAP projection to 2D $\rightarrow$ S1 Normalization (Unit Circle) $\rightarrow$ Gaussian KDE.
- **What to look for:**
 - *Uniformity:* Do features cover the space?
 - *Peaks:* Distinct peaks indicate specific feature orientations.
 - *Collapse:* Overlapping distributions imply loss of modality-specific signal.

MovieLens 1M: Aligned vs Disaligned

Disaligned (MiniLM+ViT)

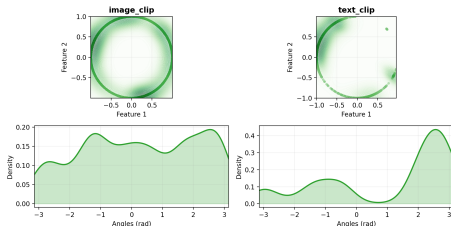
MOVIELENS_1M - text_minilm_image_vit



Distinct feature spaces preserve complementary info.

CLIP Aligned (Text+Image)

MOVIELENS_1M - text_clip_image_clip



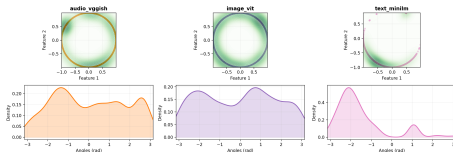
Semantic alignment bridges modalities without collapse.

- **Disaligned:** Features maintain specific orientations, capturing different aspects of user preference (e.g., visual style vs. narrative).
- **CLIP Aligned:** Successfully brings Text and Image closer semantically while maintaining enough structural diversity to be discriminative (Best Performance).

The "AudioCLIP Anomaly"

Disaligned (Rich Signal)

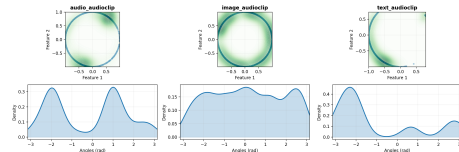
MOVIELENS_1M - text_minilm_image_vit_audio_vggish



Modality-specific distributions cover the manifold.

AudioCLIP (Over-Aligned)

MOVIELENS_1M - audioclip_full



Aggressive alignment.

Why does alignment fail here?

- **Loss of Nuance:** By forcing all three modalities (Text, Image, Audio) into the exact same space too aggressively, the model loses the rich multimodal signal

Quantitative Results: MovieLens 1M (k=10)

VBPR with CLIP features generally outperforms others. AudioCLIP underperforms.

Model	Rec@10	NDCG@10	Prec@10	MAP@10
BPR	0.1776	0.2159	0.1631	0.1120
VBPR (CLIP)	0.1911	0.2300	0.1745	0.1211
VBPR (AudioCLIP)	0.1570	0.1922	0.1476	0.0969
VBPR (Disaligned)	0.1876	0.2271	0.1718	0.1195
LATTICE_3 (CLIP)	0.1877	0.2275	0.1721	0.1198
LATTICE_3 (Disaligned)	0.1848	0.2257	0.1710	0.1183
FREEDOM (CLIP)	0.1796	0.2211	0.1657	0.1158
FREEDOM (AudioCLIP)	0.1844	0.2259	0.1688	0.1193

Table: MovieLens 1M results. **VBPR (CLIP)** achieves state-of-the-art.

Quantitative Results: LastFM (k=10)

In music domain, disaligned features preserve distinct signals better.

Model	Rec@10	NDCG@10	Prec@10	MAP@10
BPR	0.1132	0.1534	0.1404	0.1061
VBPR (CLIP)	0.1222	0.1646	0.1488	0.1149
VBPR (Disaligned)	0.1307	0.1755	0.1560	0.1242
LATTICE_3 (CLIP)	0.1232	0.1667	0.1507	0.1160
LATTICE_3 (Disaligned)	0.1258	0.1722	0.1535	0.1209
FREEDOM (Disaligned)	0.0986	0.1322	0.1224	0.0954

Table: LastFM results. **VBPR (Disaligned)** outperforms aligned models.

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Discussion: Context Matters

MovieLens 1M (Movies)

- Modalities (Posters, Plot) share high semantic correlation.
- **Result:** Alignment (CLIP) helps. It reinforces the shared signal.

LastFM (Music)

- Modalities (Audio, Cover Art) are semantically distinct/orthogonal.
- **Result:** Strict alignment hurts. Disaligned features (keeping specific audio/visual signals separate) perform better.

Conclusions and Future Work

- **Aggressive Alignment (AudioCLIP):** Can be detrimental if it collapses the feature space too much, erasing fine-grained details needed for ranking.
- **Simplicity Wins:** Simple projection models (VBPR) with high-quality features often beat complex GNNs (LATTICE) on these datasets.
- **Future Directions:**
 - Develop domain-adaptive fusion strategies (soft-alignment).
 - Investigate regularization techniques to prevent feature collapse in tri-modal alignment.

Thank you!

Thank you for your attention!



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